"CHOPPER"
a PELLETIZER by METAFUSE

Author Alvin Tanto (MateriumLabs)
alvin@3dprint.id
alvin@metafuse.tech
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OFF THE SHELF PARTS

Parts kit available at www.metafuse.tech

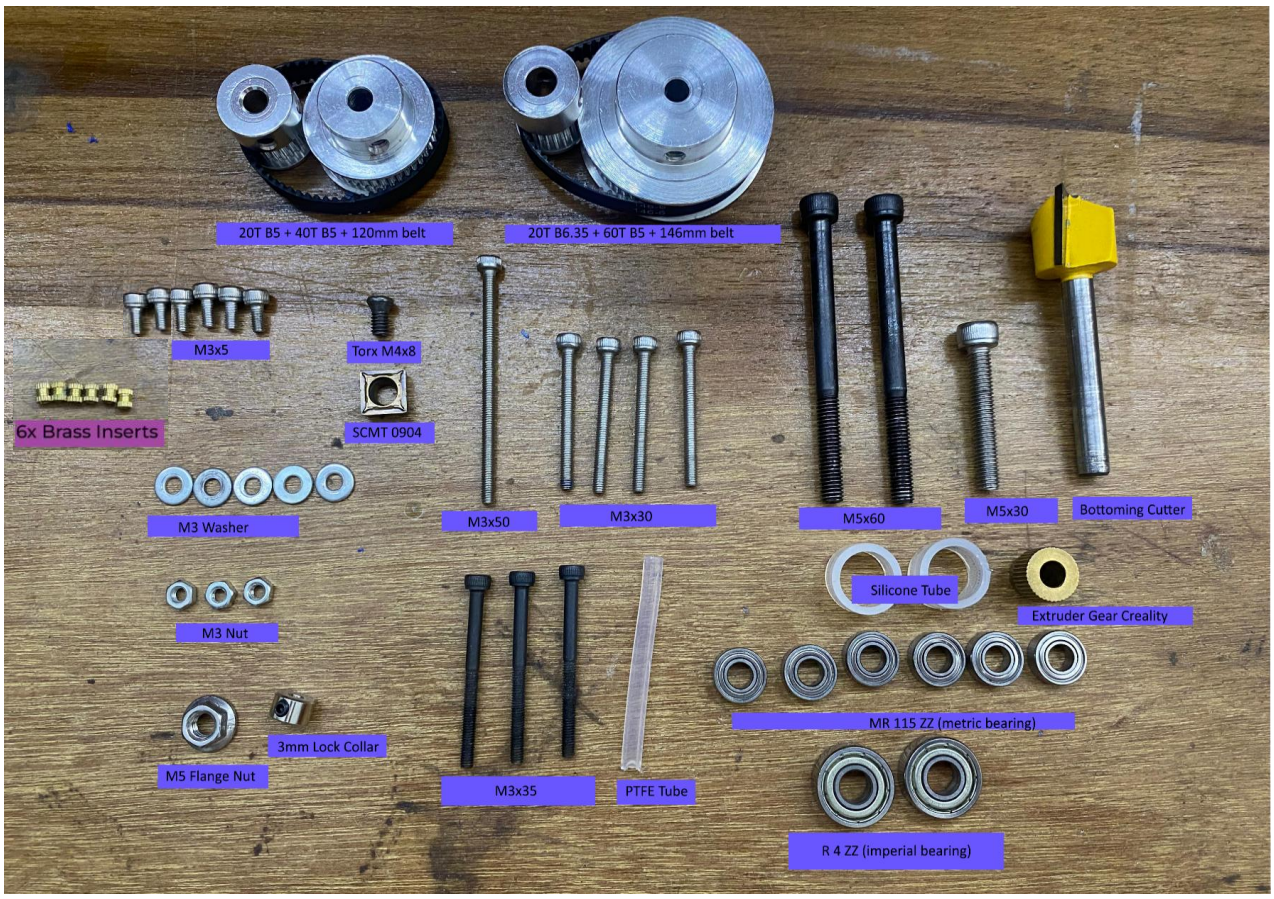
Category	Model	Quantity	Note
Hex Screws	M3x5 SHCS	6	
	M3x50 SHCS	1	Full thread
	M3x30 SHCS	4	
	M3x35 SHCS	3	
	M5x30 SHCS	1	standard 0.8mm pitch
	M5x60 SHCS	2	
Insert Screw	M4x8 Countersunk T10	1	Screws to hold cutter inserts in CNC/Lathe Machining, uses Torx Keys
Hex Nuts	M3	3	Max Thickness : 2.5mm
	M3 Washer	5	Max OD: 8mm
	M5 Flange Nut Serrated	1	Replacable with Normal M5 Hex Nut
	M5 Shim	1	Optional, Max OD : 10mm
Stationary Knife	SCMT 0904	1	SCMT 0904 carbide insert (standard machining insert)
Bearings	MR115ZZ	6	5×11×4mm (commonly used in RC cars / 1:28 scale)
	R4ZZ	2	1/4" × 5/8" × 0.196" (6.35×15.875×5mm)
Transmission	GT2-60T-B5-W6 Pulley	1	Hole 5mm, Belt Width 6mm
	GT2-40T-B5-W6 Pulley	1	
	GT2-20T-B5-W6 Pulley	1	
	GT2-20T-B6.35-W6 Pulley	1	Hole 6.35mm, Belt Width 6mm
	GT2-120-W6 Closed Loop Belt	1	120mm loop, 60 Tooth
	GT2-146-W6 Closed Loop Belt	1	146mm loop, 73 Tooth
	3mm Shaft Collar Lock	1	3mm ID, 8mm OD, 5mm Thickness
	Creality Extruder Gear	1	MK8-style drive gear for Creality extruders (5 mm bore, ~11 mm OD)
	PTFE Tube	1	2mm ID, 4mm OD, 50mm Length
	8mm to 6.35mm Shaft Coupling Rigid	1	
	Silicone Tube	2	11mm ID, 14mm OD, 8mm Length (cut by yourself from stock)
Rotary Knife	19 mm bottom-cut router bit	1	min. Shank length 40mm or more, Shank dia.6,35mm
Heat-set Inserts	Brass Insert M3x5mm	6	Straight or twisted twills is fine
Tools	Torx T10	1	Torx T10 Key

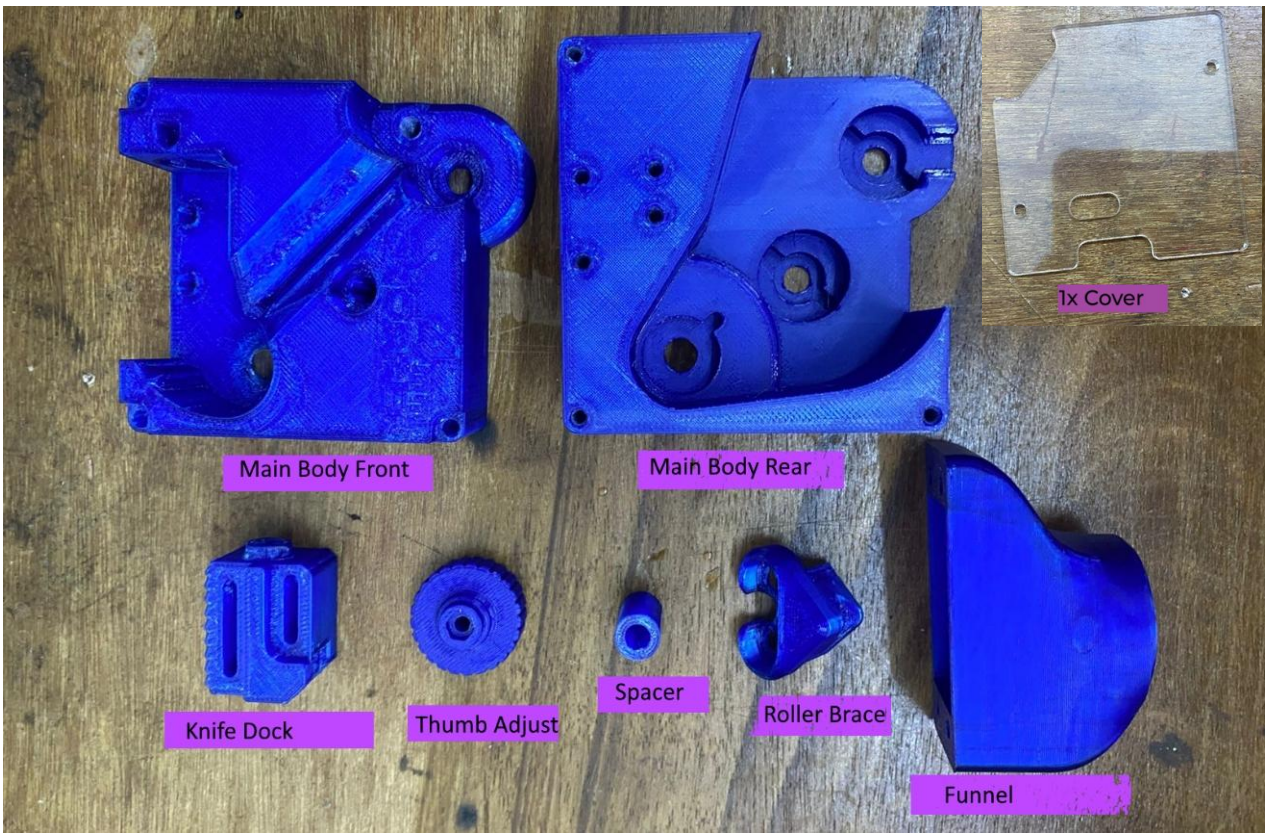
3D PRINTED PARTS

Parts kit available at www.metafuse.tech

Category	Material	Quantity	Recommended Print Parameter
Main Body Front	PETG	1	5 Outer Shell, 25% Triangular Infill, 0,2mm Layer Height. Printed parts must be within ±0.1 mm dimensional accuracy. Otherwise the belts may become too loose or too tight to assemble. Print orientation must follow guide.
Main Body Rear	PETG	1	
Knife Dock	PETG	1	
Thumb Adjust	PETG/PLA	1	
Roller Brace	PETG/PLA	1	
Funnel	PETG/PLA	1	
Spacer	PETG/PLA	1	
Cover	Acrylic 3mm or 3D Printed	1	

REFERENCE PICTURE





TOOLS REQUIRED
Allen Key for M2, M3, and M5 screws
Torx T10 key
M3 Tap
Pliers
Soldering tool or heat insert tool
Box Cutter to cut Tubes
M4 Tap (optional)
M5 Tap (optional)
M5 Spanner (optional)
Tweezers (optional)